

A series of white, overlapping geometric lines and polygons on a black background, creating a complex, abstract pattern that resembles a stylized map or a network of paths.

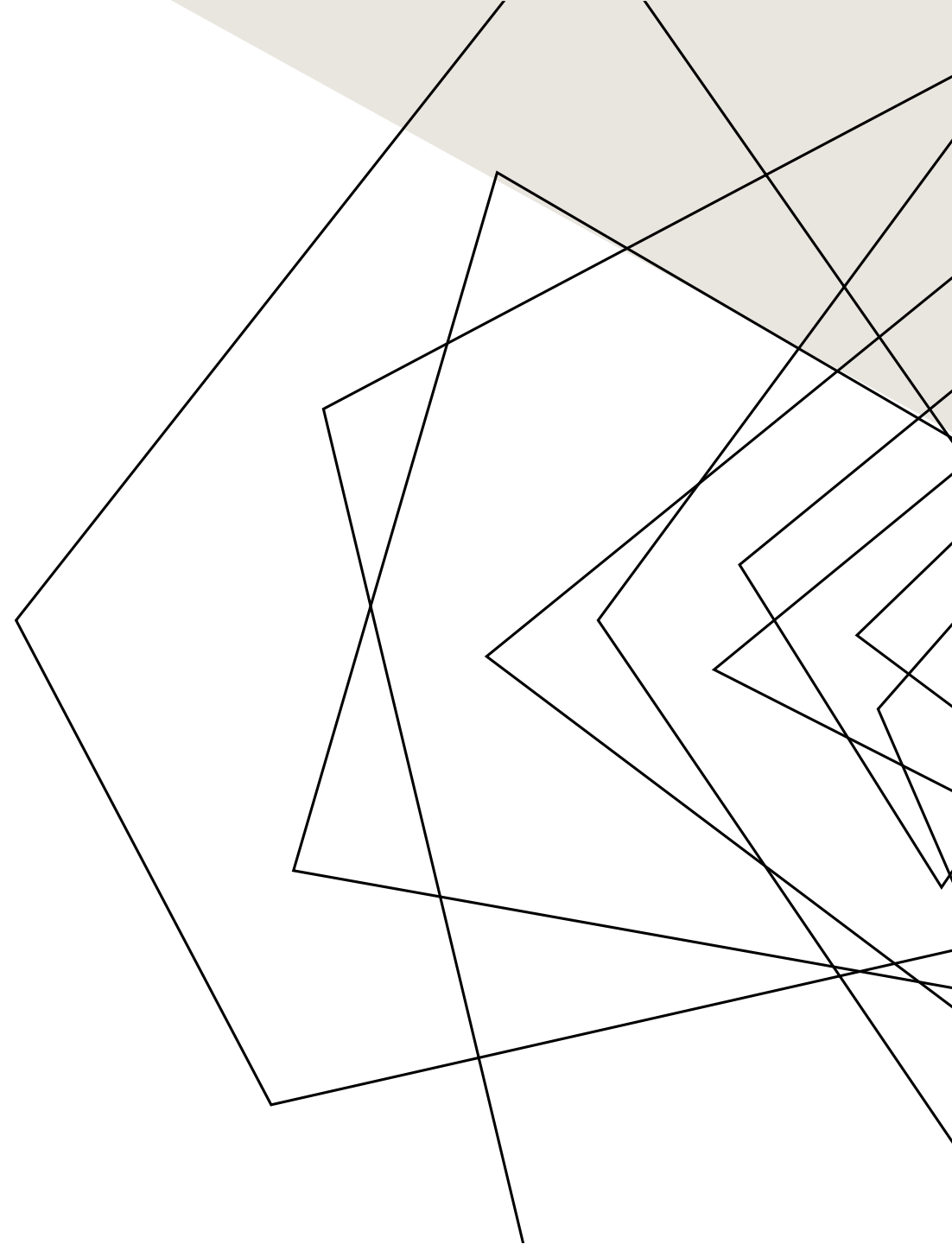
NUMERICAL PROGRAMMING FINAL PROJECT RETAKE

Path Following and Navigation in Dynamic Environments

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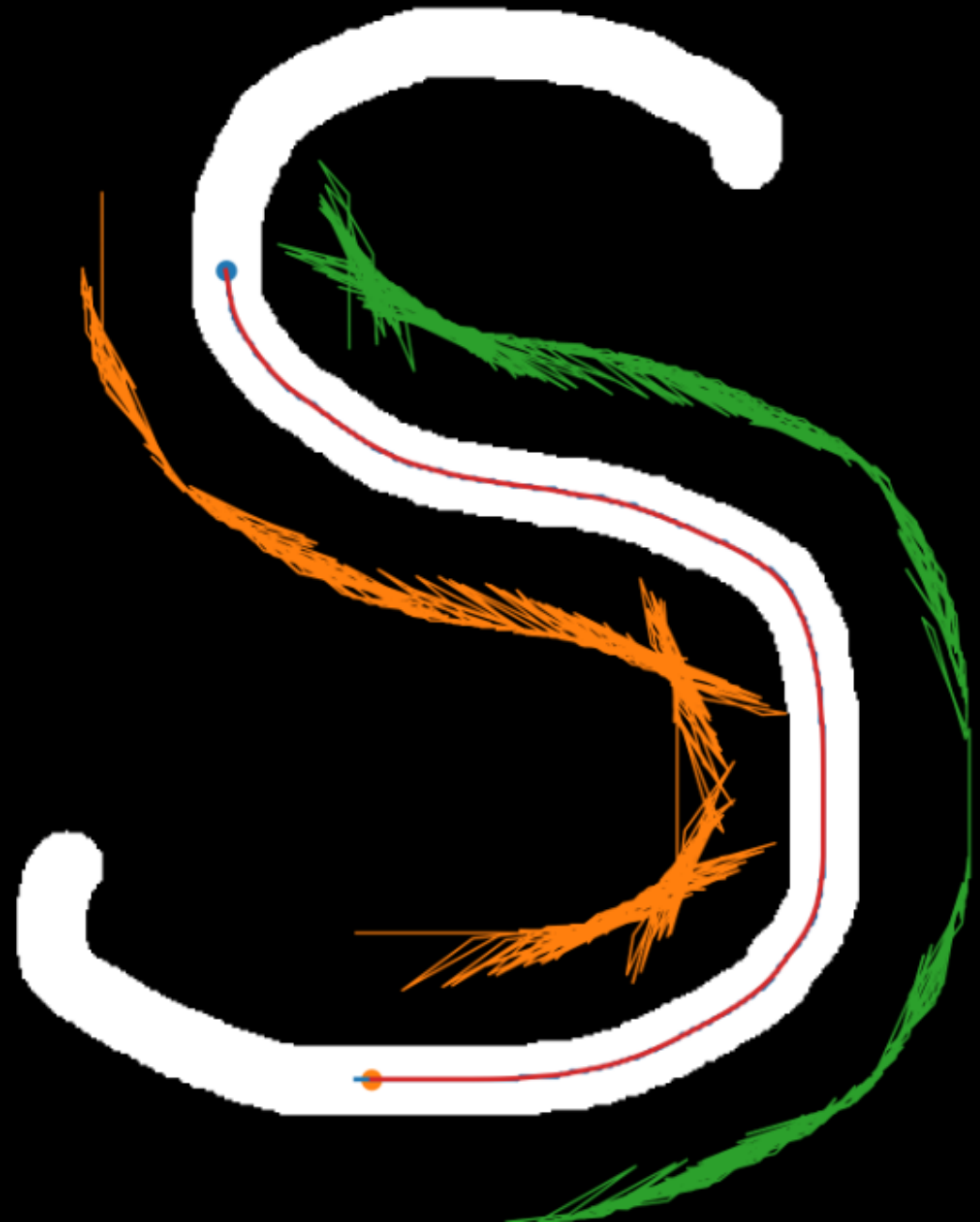
PROJECT OVERVIEW

- Task 1 – Path extraction & tracking
- Task 2 – Bidirectional swarm navigation
- Task 3 – Robot navigation in pedestrian flow
- Automated testing & limitations



TASK 1: PATH EXTRACTION

- Binary map $\rightarrow$ skeletonization
- BFS shortest path
- Spline fitting
- Corridor construction



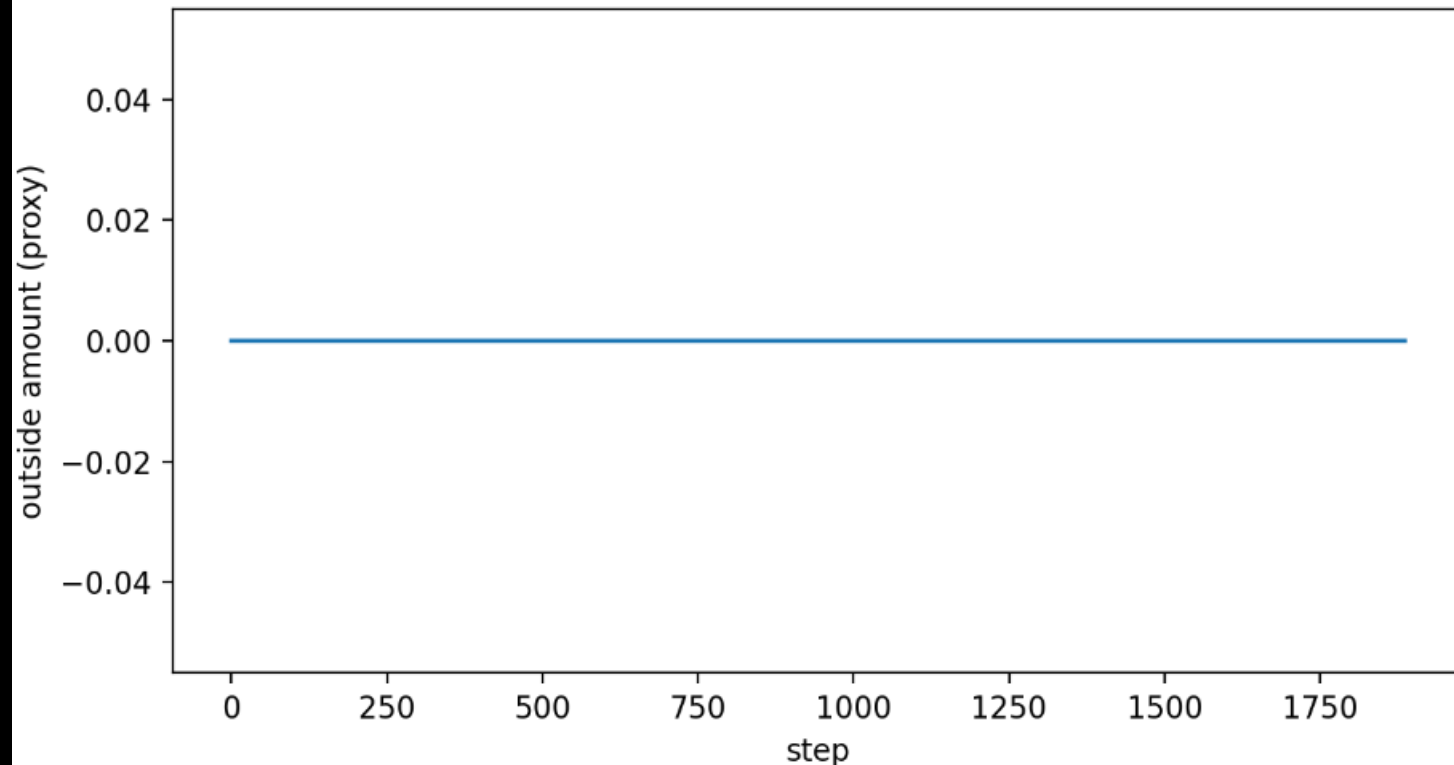
ROBOT TRACKING MODEL

- Look-ahead tracking
- Border penalty keeps robot inside corridor

$$\dot{x} = v$$

$$\dot{v} = \frac{1}{m}(k_p(T - x) - k_d v + F_{border})$$

Task 1: Border violation indicator (0 = inside)



Task 2: Bidirectional swarms ($R_{\text{safe}}=36.0\text{px}$)



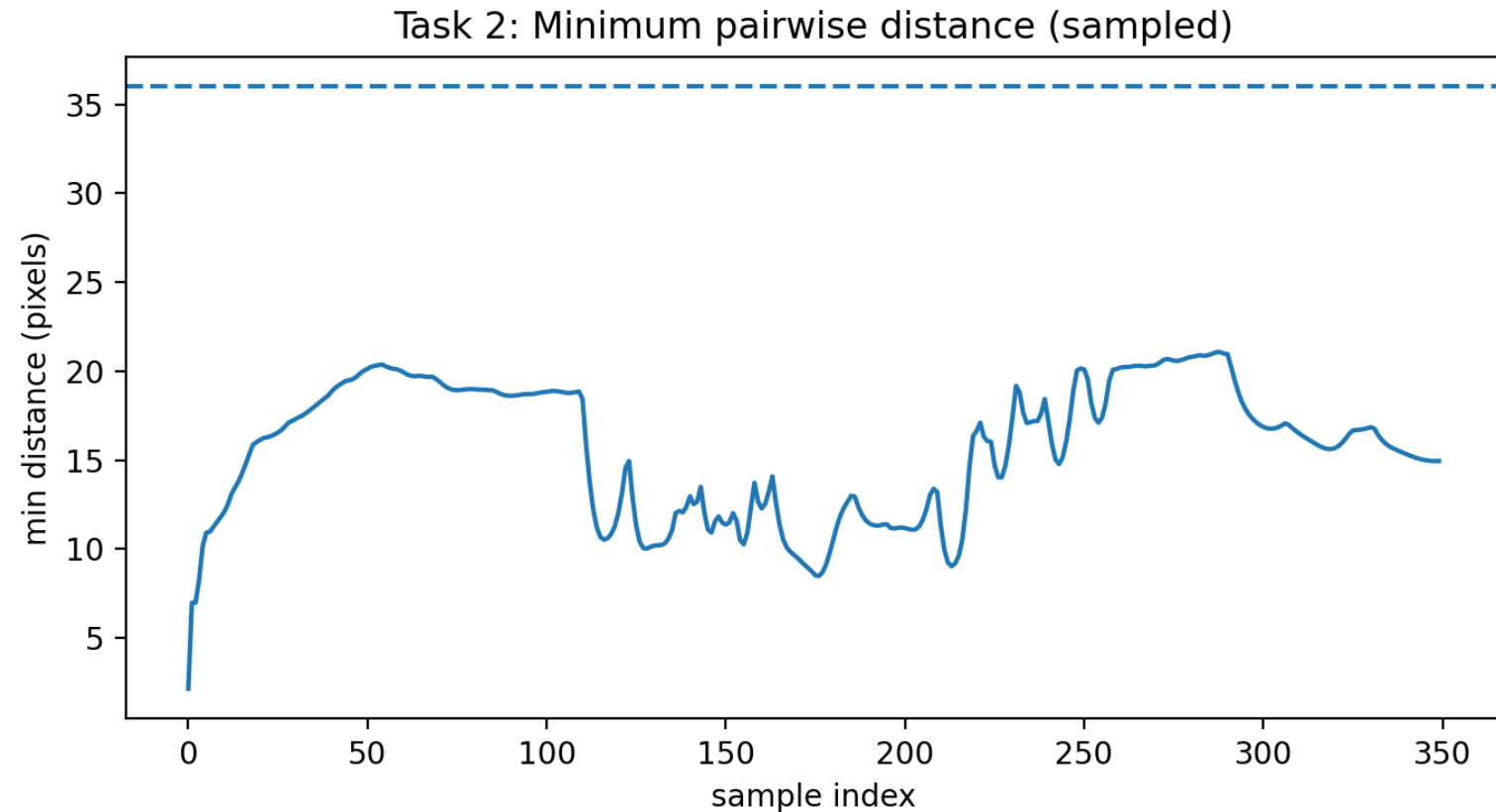
TASK 2: BIDIRECTIONAL SWARMS

- Two groups A and B
- Opposite directions
- Lane offset
- Smooth repulsion

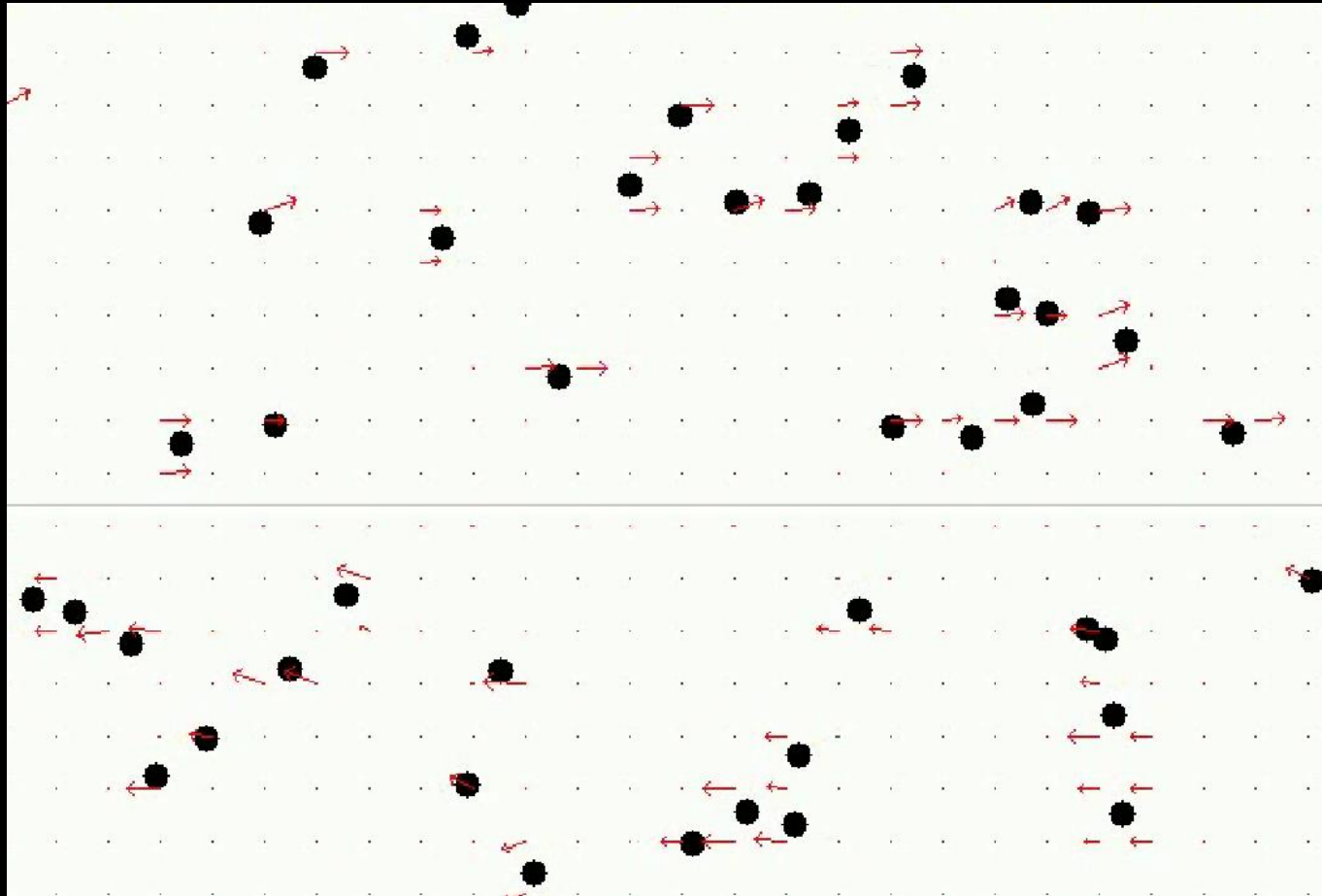
$$F_{\text{rep}}(d) = \begin{cases} k_{\text{rep}} \left(\frac{1}{d} - \frac{1}{R_{\text{safe}}} \right) \hat{r}, & d < R_{\text{safe}}, \\ 0, & d \geq R_{\text{safe}}, \end{cases}$$

COLLISION AVOIDANCE PERFORMANCE

- Safety threshold:
 R_{safe}
- Minimum distance over time
- Works under tuned parameters



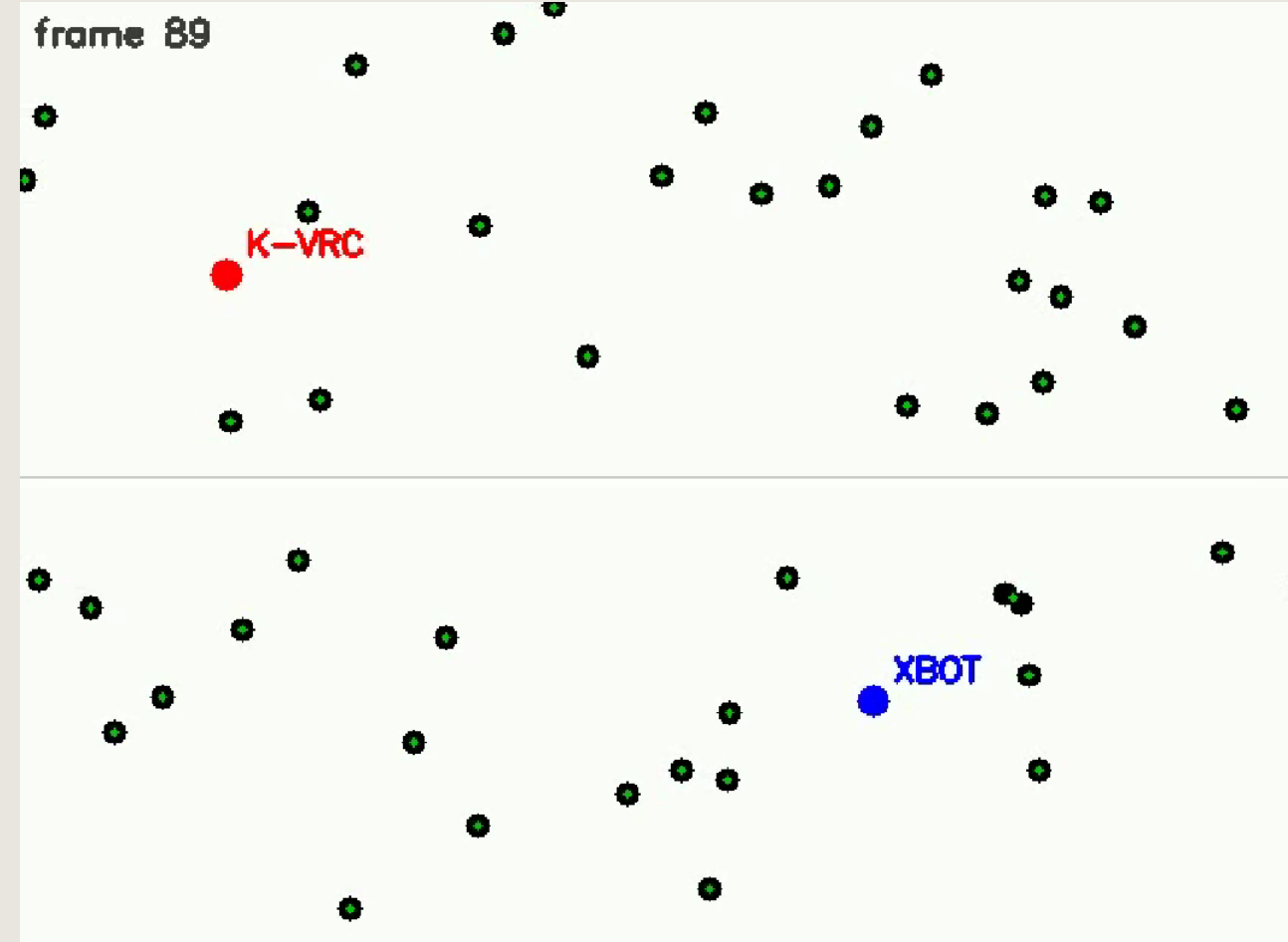
OPTICAL FLOW BASED VELOCITY FIELD



- Synthetic pedestrian video
- Two opposite flows
- Farnebäck optical flow

NAVIGATION IN DYNAMIC CROWD

- K-VRC (top flow)
- XBOT (bottom flow)
- Flow tracking
- Repulsion from pedestrians
- Direction switching

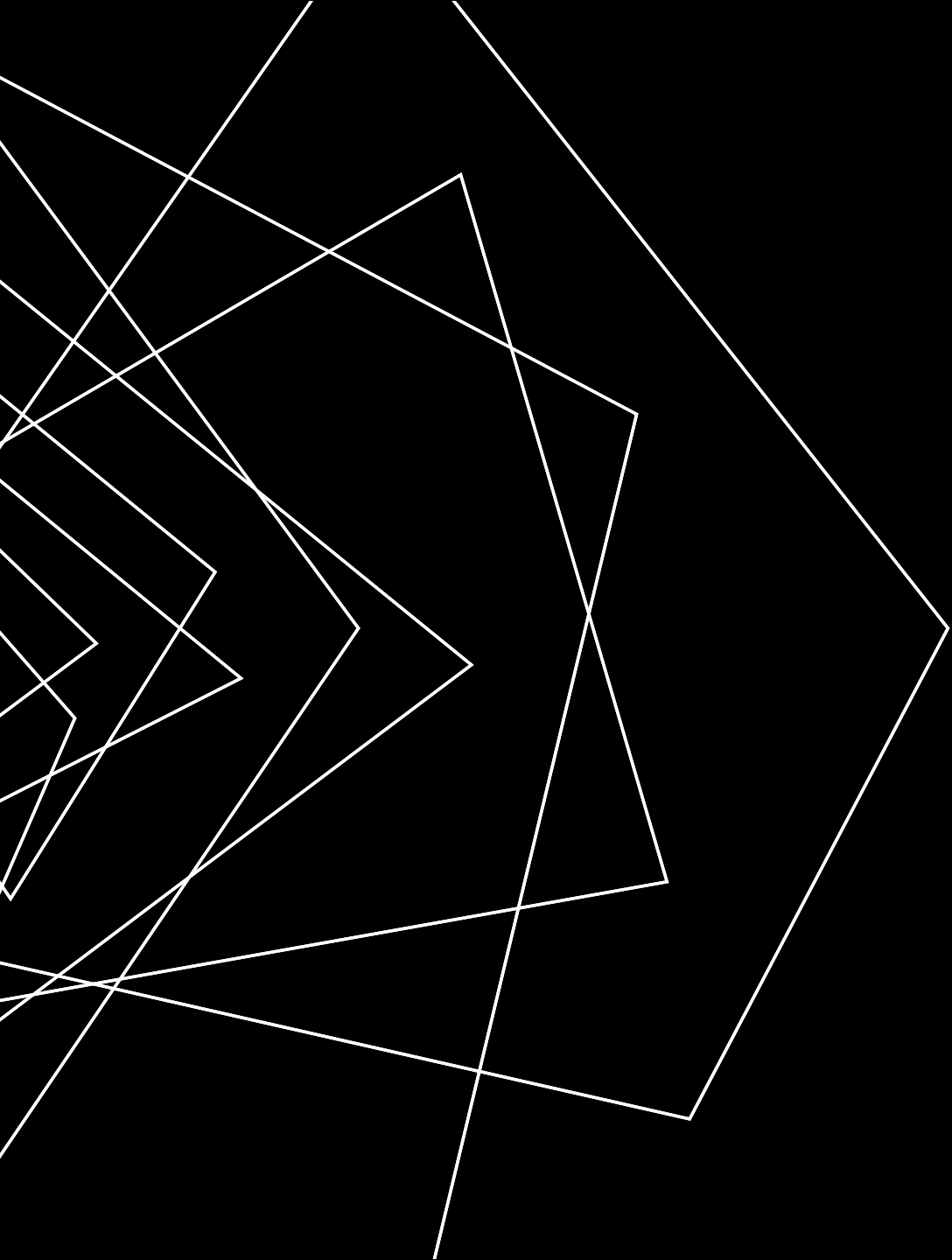


TESTING AND LIMITATIONS

- ✓ RK4 convergence
validated
- ✓ Path tracking
validated
- ✓ Optical flow direction
validated
- ⚠ Swarm safety depends
on density and tuning

collected 6 items

```
tests/test_numerics_convergence.py::test_rk4_convergence_harmonic_oscillator PASSED [ 16%]  
tests/test_task1_following.py::test_task1_following_inside_corridor PASSED [ 33%]  
tests/test_task1_following.py::test_task1_fails_when_corridor_too_narrow XPASS (Limi...) [ 50%]  
tests/test_task2_bidirectional.py::test_task2_bidirectional_progress PASSED [ 66%]  
tests/test_task2_no_collision.py::test_task2_no_collision_basic FAILED [ 83%]  
tests/test_task3_flow_follow.py::test_task3_optical_flow_two_opposite_flows PASSED [100%]
```



CONCLUSION

- Unified modeling approach
- Reproducible via YAML configs
- Modular architecture
- Demonstrated working and failure cases

THANK YOU